

Lab 11 Activity

We will work with the same dataset that we used in the main lab. You can read in the data using the following code:

```
library(tidyverse)
library(haven)
dat <- read_dta("Lab_11_attend.dta")
```

1

The variables `frosh` and `soph` are dummy variables for the student being a freshman or sophomore (0 on both variables indicates 'other'). Fit a model predicting final exam scores from (the centered) prior GPA with academic year as a moderator (no need to modify the categorical moderator variables). Is there a statistically significant moderation effect?

2

Construct a plot to illustrate the moderation effect (use separate lines for different grade levels). How does this plot help you interpret your results from step 1?

3

Using this same data set, conduct a mediation analysis predicting final exam scores from ACT scores with `hwrt` (rate of completing homework) as a mediator (you may drop the 6 missing values of `hwrt` for this step). What do you conclude?

4

Repeat #3 using a method that using a better method of missing data handling than listwise deletion.